Team Name : BAH BAH black sheep have you any RAM

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Problem Statement : Air-Gapped Predictive Copilot for Secure MPLS Operations

APPROACH AND NOVELTY

THE GAP

Reactive-only alerting
Tools fire alerts after threshold breach, not before. No lead time to intervene before user-visible degradation.

Buried precursor signals
Congestion buildup, latency drift, jitter trends, tunnel decay all sit unseen in raw SNMP/syslog/BGP/NetFlow data. Manual correlation is slow, error-prone, drives alert fatigue.

Air-gap lockout
Regulated/government networks ban cloud-connected AI. The highest stakes environments get zero intelligent guidance.

THE SOLUTION

Predictive analytics engine
A fusion of anomaly detection models forecast congestion, routing instability, tunnel degradation ahead of impact, with time-to-impact estimates.

Offline LLM copilot
Fully self-hosted LLM + local RAG over topology, runbooks, incident history. Explains predicted issue, root cause, confidence, and next action in plain language- zero outbound network calls.

Operator answers, in real time
What fails next and when. Why risk is elevated. What action to take before SLA/security impact - all within the air-gapped boundary.

OUR NOVELTIES

1 Predictive Analytics System

Simulation-grounded accuracy: Trained on real fault labels + known time to impact from the topology itself.
Meta-learning: Meta-learner picks best base model per fault and adapts fast to new variants.
Multi-signal fusion: SNMP + BGP/OSPF + tunnel + NetFlow fused per device, precursors read across pillars.

2 Copilot

Artifact sandbox: Read-only, cited retrieval over topology/incidents/runbooks. No hallucination, no write access. Protected code execution environment for AI copilot to generate graphs and other artifacts
Topology-aware intelligence: Root-cause reasoning grounded in live topology graph, includes real blast radius.

3 Shadow Deployment

New models run shadow alongside production on live/replayed traffic, no operator exposure. Promoted only if it beats baseline on accuracy, lead time, false-positive rate.

CORE CAPABILITIES

Realistic Simulation

- 148-container Containerlab topology (24 P-routers/6 POPs, 12 PE route-reflectors, 34 CE routers across branch/hub/DC, 78 per-VRF hosts), real FRR BGP/OSPF, BFD 300ms, WireGuard SD-WAN, netem/HTB impairment, D-ITG dynamic load (40–95% over 24h diurnal cycle).
- 21 injectable fault scenarios that cascade through the topology rather than staying local and logs important parameters

Grafana Dashboard

The NOC Overview dashboard: 11 panels covering interface utilization, OSPF adjacency/SPF, tunnel health, and BGP state. A custom Grafana plugin surfaces the topology map, fault injector, predictive alerts, and copilot chat directly inside the operator's existing dashboard no switching to a separate tool. Built on React JS, it is fully customizable & extensible; Grafana itself is the industry-standard NOC visualization platform, meaning operators work in a familiar environment while the AI layer integrates seamlessly underneath.

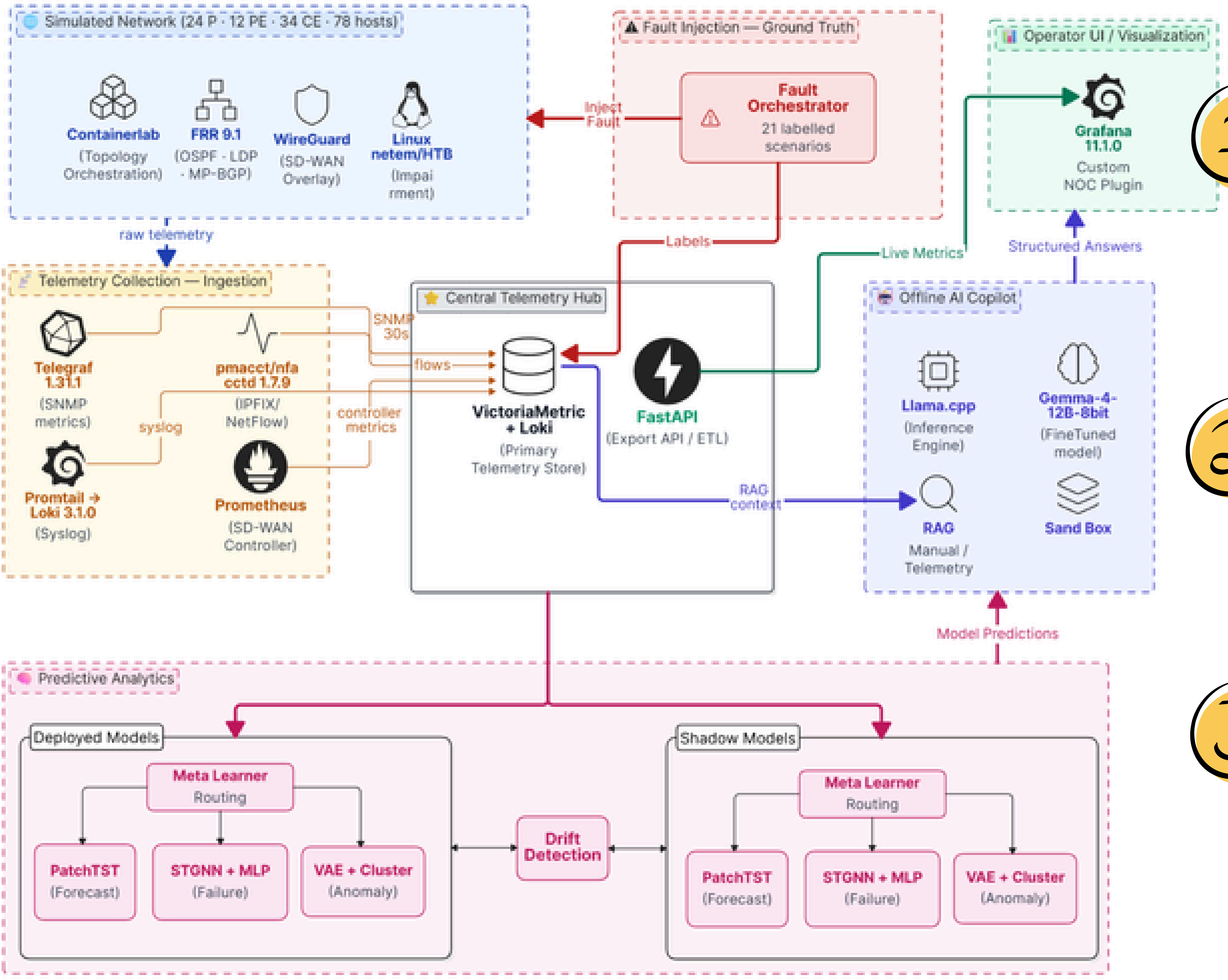
RAG Stack

Three retrieval surfaces feed the copilot's context window: RAG over logs (syslog/BGP-OSPF event history in Loki, for "what happened when"), RAG over docs (runbooks + incident write-ups in ragcorpus/, for remediation guidance), and KG over network topology (graph-structured device/link/VRF relationships, for blast-radius and root-cause reasoning), all local, zero outbound calls.

Topology Aware Copilot

- STGNN graph attention weights are intrinsic, not bolted-on. Every prediction includes contributing devices/links, feature importance, root cause ranking, and nearest past incident from memory.
- The network topology is represented as a knowledge graph where nodes are devices and encode faults, edges encode protocols and capture link quality, giving the copilot LLM a queryable map of the network so it can reason about failures in topological context.

ARCHITECTURE DIAGRAM



OUR NOVELTIES

1 Telemetry Ingestion

The simulated network (Containerlab · FRR · WireGuard) streams raw SNMP, syslog, NetFlow/IPFIX, and SD-WAN controller metrics through the collection stack (Telegraf, Promtail, pmacct, Prometheus) into the Central Telemetry Hub (VictoriaMetrics + Loki), which normalizes everything into a single time-series store. The Fault Orchestrator injects labelled faults in parallel, feeding ground-truth labels alongside the live telemetry.

2 Predictive Analytics

The Hub fans telemetry out to dual Deployed + Shadow model banks, each routing signals through a Meta-Learner to PatchTST (forecast), STGNN+MLP (failure), and VAE+Cluster (anomaly); Drift Detection compares the two banks to trigger safe promotion. Outputs are time-to-impact predictions with contributing-signal attribution.

3 Copilot & Operator UI

Model predictions plus RAG context (topology, runbooks, past incidents) are pushed via FastAPI into the offline LLM (Llama.cpp + fine-tuned Gemma), which returns structured, grounded answers- predicted issue, confidence, root cause, affected scope, action rendered in the Grafana NOC plugin. All flows stay inside the air-gap boundary with zero outbound traffic.

USER EXPERIENCE

User Experience Enhancers

Guided AI Troubleshooting

Provides intelligent, step-by-step diagnostics by analyzing live telemetry, topology, and historical incidents. Explains probable root causes, recommends prioritized remediation actions, and helps operators resolve issues quickly without extensive manual investigation.

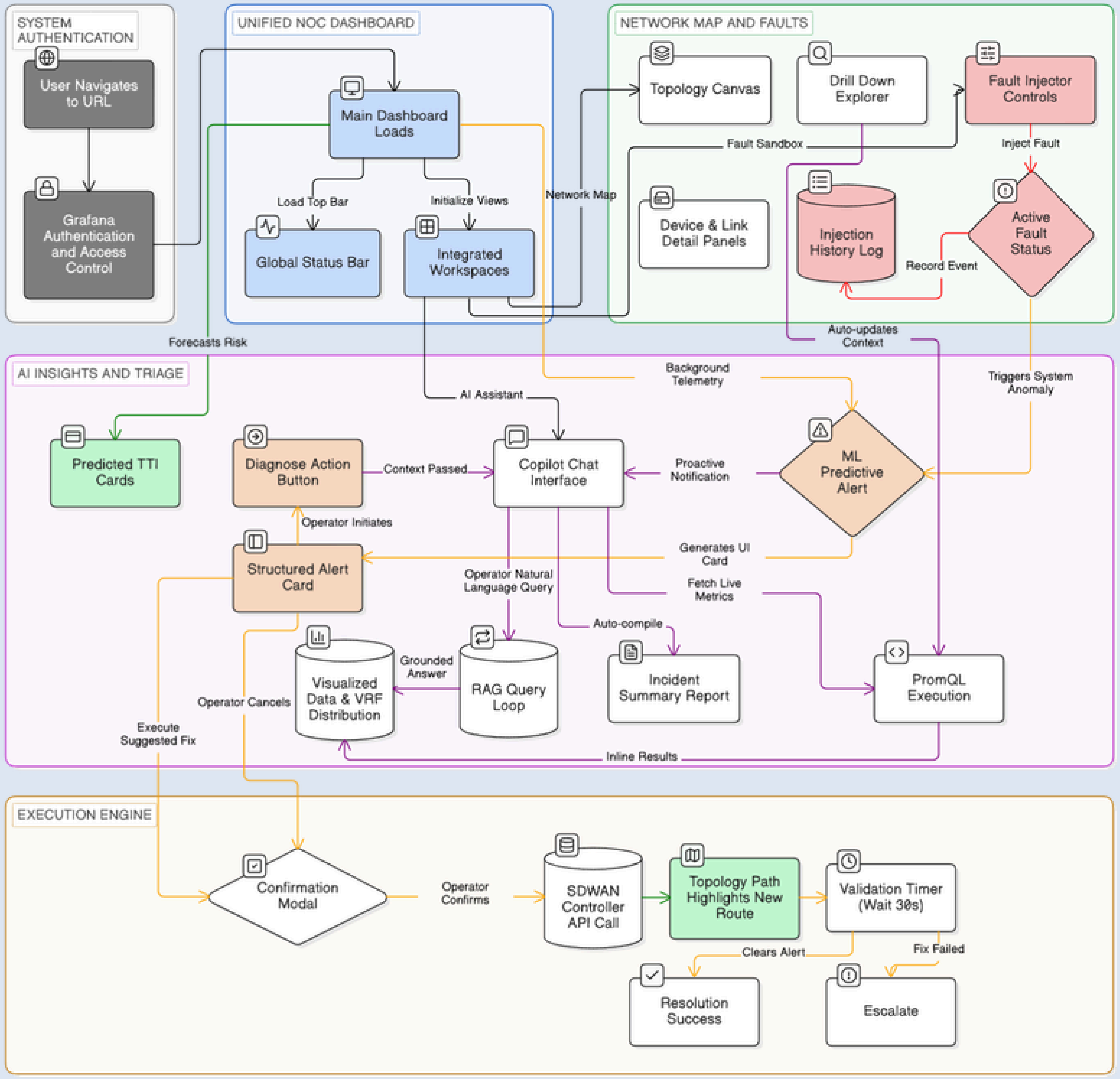
Intelligent Alert Prioritization

Ranks alerts based on confidence, predicted impact, and estimated time-to-failure instead of simple threshold breaches. Filters redundant notifications, reduces alert fatigue, and enables operators to focus on the most critical issues first.

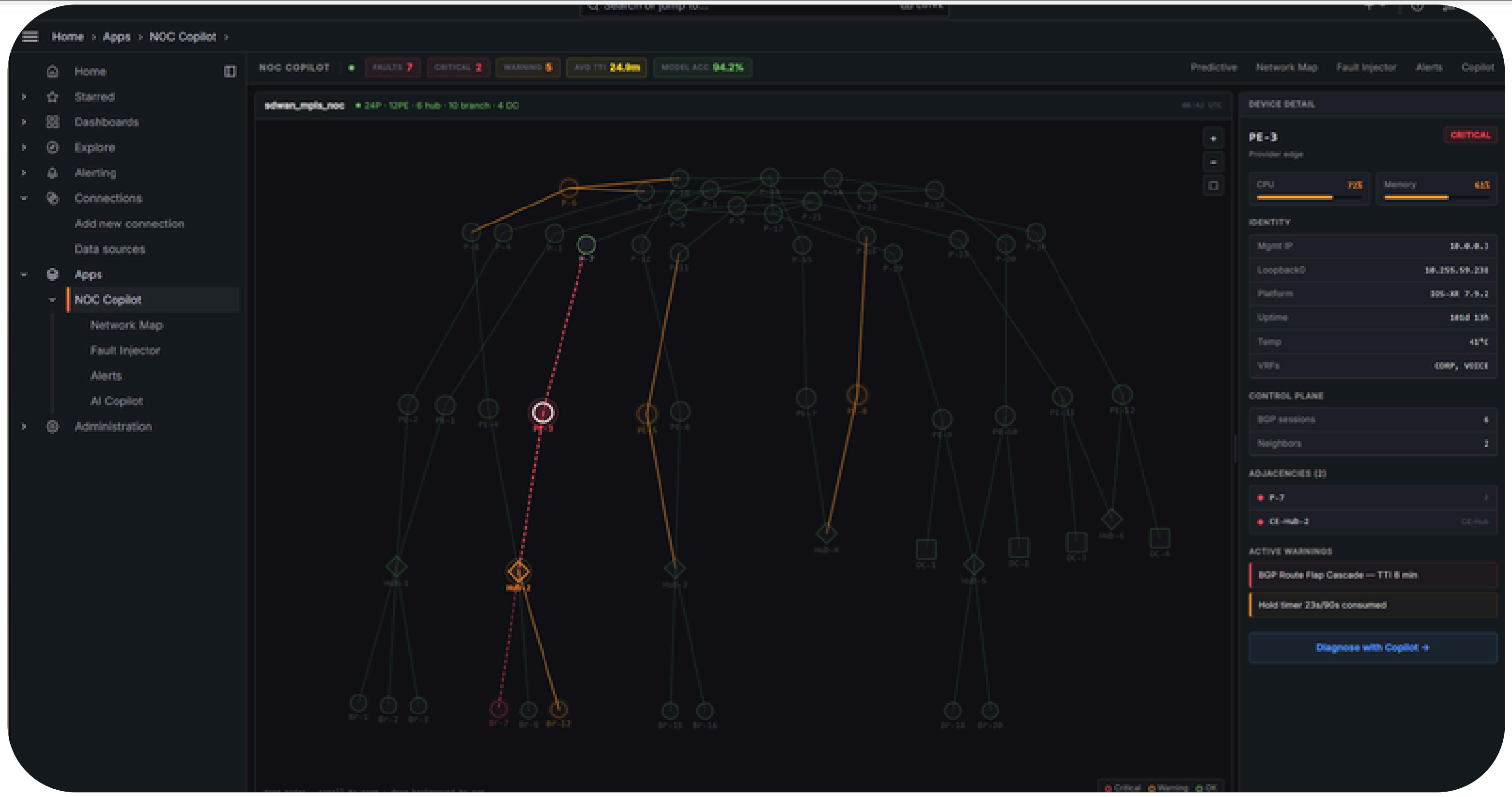
Interactive Topology Visualization

Displays a real-time, interactive network map with fault localization, dependency tracing, and one-click drill-down into affected devices, links, and services. Simplifies navigation and accelerates root-cause analysis across complex SD-WAN/MPLS environments.

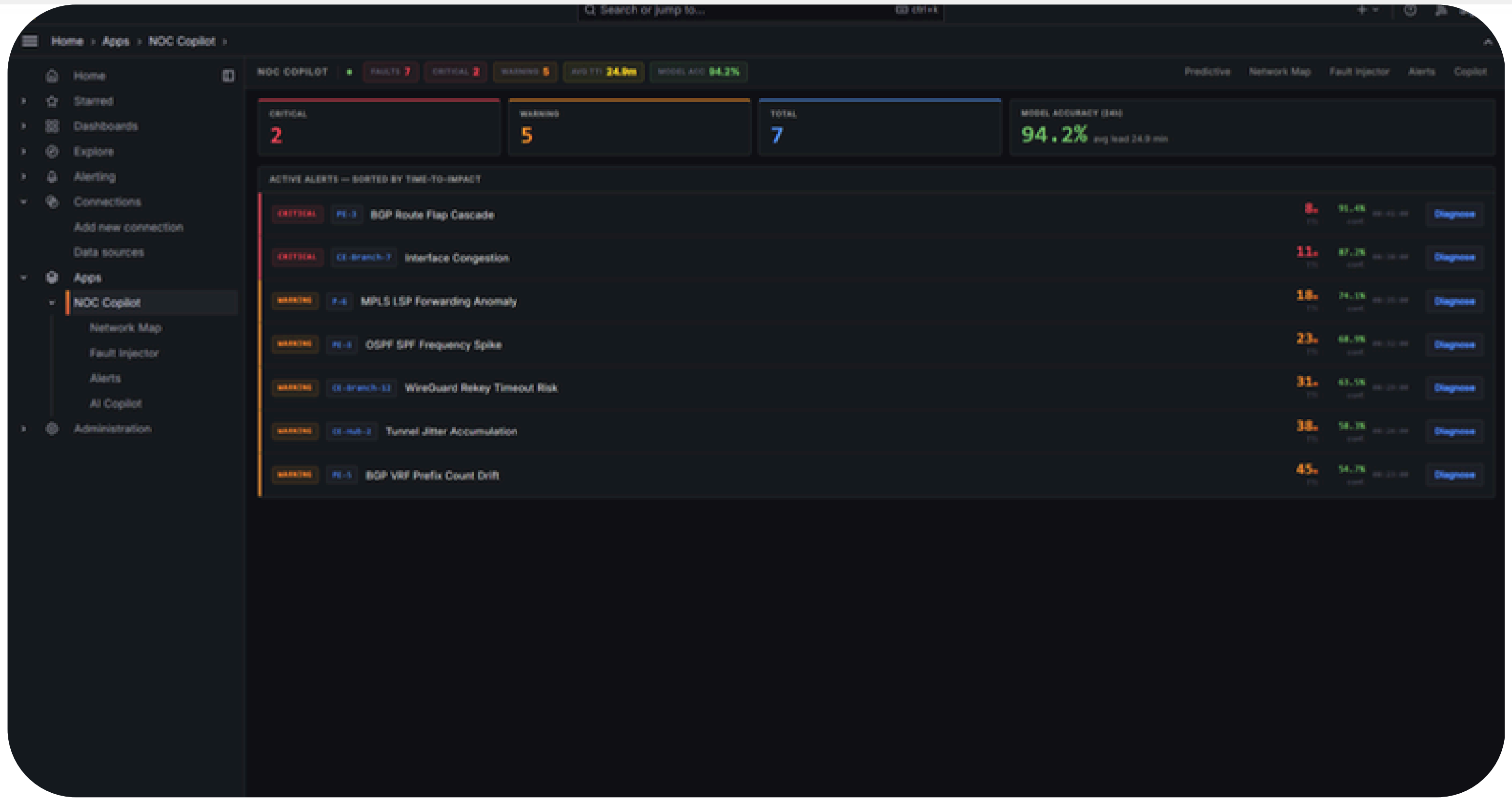
User Flow Diagram



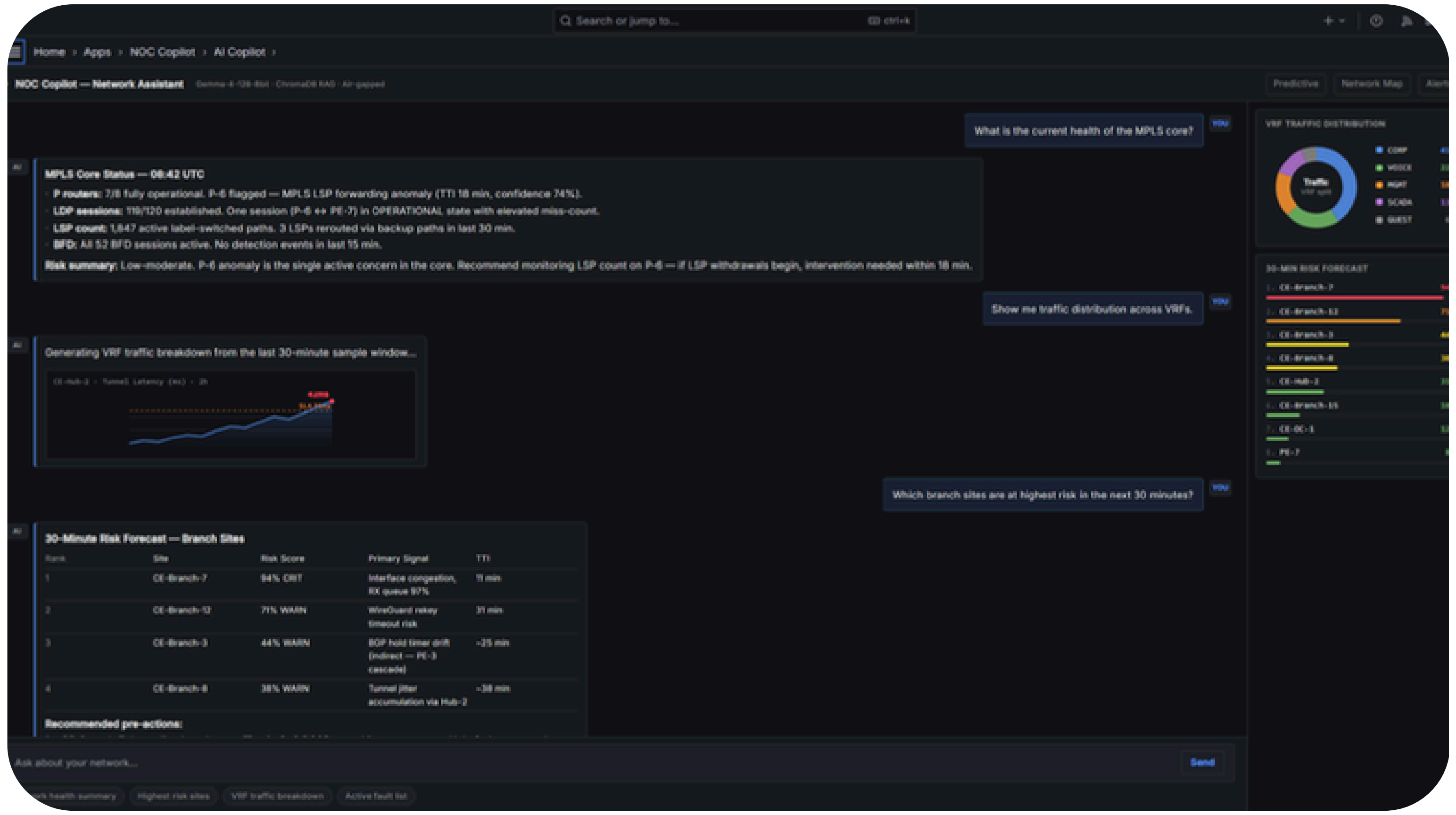
PRODUCT MOCKUPS: GRAFANA NOC PLUGIN



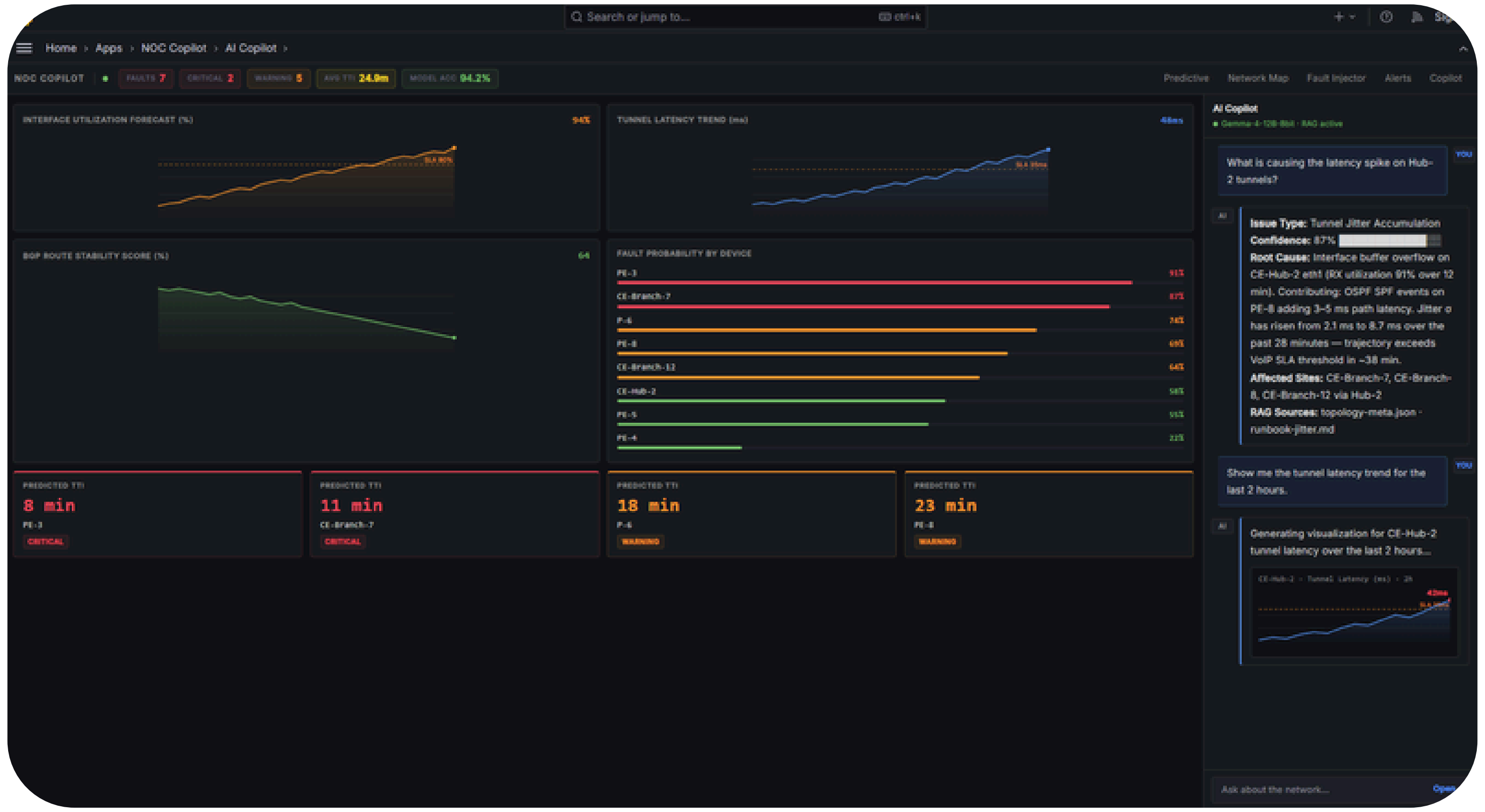
Network Map: Live topology + per-device detail & fault overlay



Alerts: Ranked by time-to-impact/severity · one-click diagnosis



AI Copilot: Offline NL assistant · RAG-cited · risk forecast



Predictive Dashboard: Forecasts · Fault-Probability · Time-to-Impact

Check our progress!

Repo

TECHNOLOGIES USED

Network Simulation: Containerlab, FRR 9.1, WireGuard, Linux netem

Telemetry Collection: Telegraf, pmacct, Promtail, Loki, Prometheus

Central system: VictoriaMetrics, Fast API, Grafana,React, TypeScript, Webpack

Offline AI Copilot: Llama.cpp, Gemma-4-12B, ChromaDB, FAISS

Predictive Analytics: PyTorch, PatchTST, STGNN + MLP, VAE + Cluster, NumPy, pandas, and scikit-learn

scikit learn

CONTAINERlab

Research Papers

1. PatchTST: A Time Series is Worth 64 Words (Long-Term Forecasting)
2. Graph WaveNet for Deep Spatial-Temporal Graph Modeling
3. Deep Autoencoding Gaussian Mixture Model for Unsupervised Anomaly Detection
4. DOTE: Rethinking Predictive WAN Traffic Engineering
5. Tiresias: Online Anomaly Detection for Hierarchical Operational Network Data
6. Spatio-Temporal Graph Convolutional NetworksHeterogeneous Graph Neural Network

ESTIMATED COST

One on-prem box runs the Copilot + telemetry stack. Concurrency = simultaneous chat generations (vLLM batching); bursty NOC usage supports ~3-5× as many logged-in operators.

PILOT / SINGLE DESK

1–2

concurrent chats
≈ 5-8 operators

RTX 5060 Ti 16 GB
Ryzen 5 7600
32 GB · 1 TB NVMe

₹1.15 L

12B at 8-bit fills 16 GB with ~2 GB spare for KV-cache, one or two live generations. Single NOC desk / pilot.

SMALL NOC

4–6

concurrent chats
≈ 15-20 operators

RTX 5070 Ti 16 GB
Ryzen 7 7700
64 GB · 2 TB NVMe

₹1.90 L

Serve 12B at Q5 + continuous batching → VRAM frees for 4–6 parallel KV caches; faster card cuts latency.

SHIFT TEAM

10–14

concurrent chats
≈ 40 operators

RTX 5090 32 GB
Ryzen 9 / Core i9
128 GB · 2×2 TB

₹3.90 L

32 GB → 12B-Q8 (13 GB) + ~18 GB KV pool = 10-14 concurrent 4k-context streams; retrain while serving.

ENTERPRISE / MULTI-SITE

30–50

concurrent chats
100+ operators

RTX PRO 6000 96 GB
or 2× RTX 5090 32 GB
Threadripper
256 GB · NAS

from ₹6.5 L

96 GB (or dual-GPU) → large KV pool for 30–50 chats, or a bigger model + org-wide API. ECC, 24×7.

For additional information or clarification, please refer to our supplementary documentation:
https://docs.google.com/document/d/16vbsVKxpKsxvHx6HKG4xgF-NEsaYdZGXbQUdR3I3Jek/